XVIII. EEBIOMASS Virtual Workshop
BIOMASS mission Cal/Val activities

Statistically rigorous validating the BIOMASS Level-2 products across 4 million km^2 and six forest biomes in Brazil

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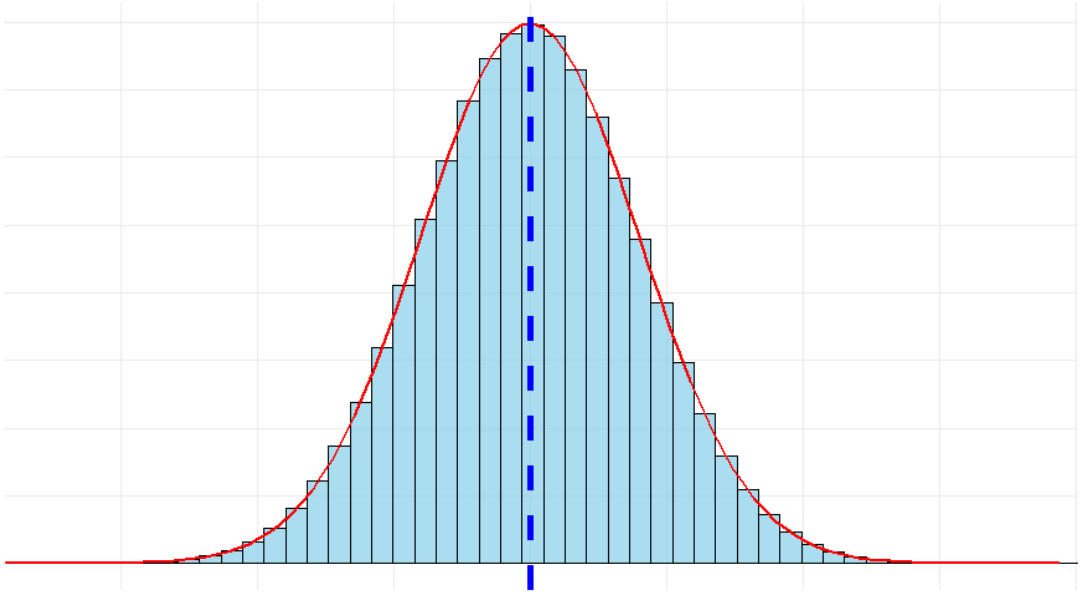
Methods Development

- Map assessment of the two main BIOMASS Level-2 products above-ground biomass (AGBD) and upper canopy height (FH):
  - Quantify and assess systematic errors in the AGBD map in various biomes in Brazil
  - Quantify and assess potential gain in precision that may be obtained when using the AGBD and FH maps as auxiliary information in design-based inference

Design-based inference

- Every population element (“pixel”) has a fixed value without error and the probability of an element being selected into the sample is known and >0

- Variability introduced through randomization in the sample selection process according to the chosen design (e.g. simple random sampling)



Estimators (design-unbiased):

- Field-based estimators:

$$\hat{Y}_{\mathrm{F}} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$\hat{V}(\hat{Y}_{\mathrm{F}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(y_i - \hat{Y}_{\mathrm{F}})^2$$

Model-assisted estimators:

$$\hat{Y}_{\mathrm{MA}} = \frac{1}{N}\sum_{i=1}^{N}\hat{y}_i + \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)$$

$$D = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)$$

$$\hat{V}(\hat{Y}_{\mathrm{MA}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}[(y_i - \hat{y}_i) - D]^2$$

$$\mathrm{RE} = \frac{\hat{V}(\hat{Y}_{\mathrm{F}})}{\hat{V}(\hat{Y}_{\mathrm{MA}})}$$

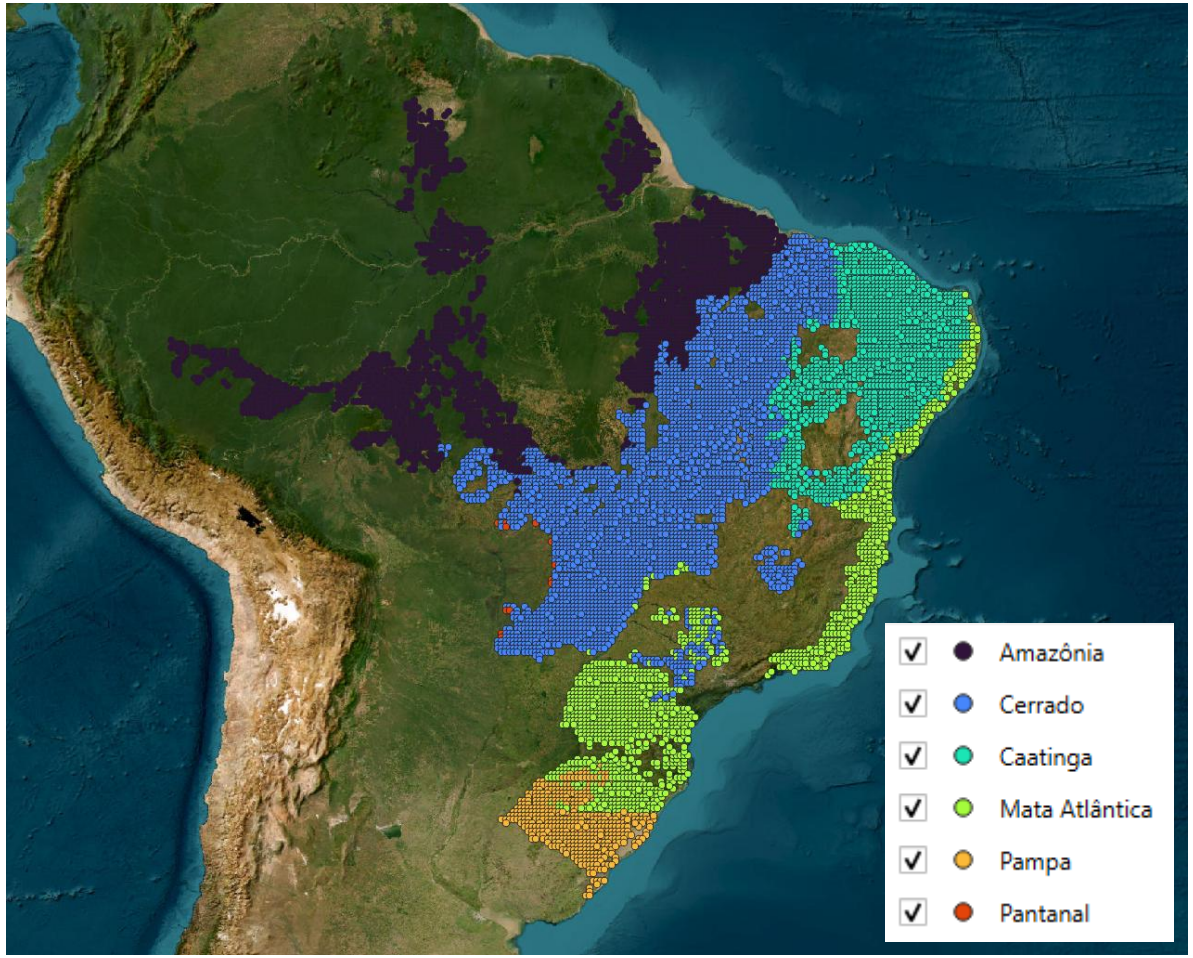
$$\mathrm{CI} = D \pm t\sqrt{\hat{V}(\hat{Y}_{\mathrm{MA}})}$$

Status

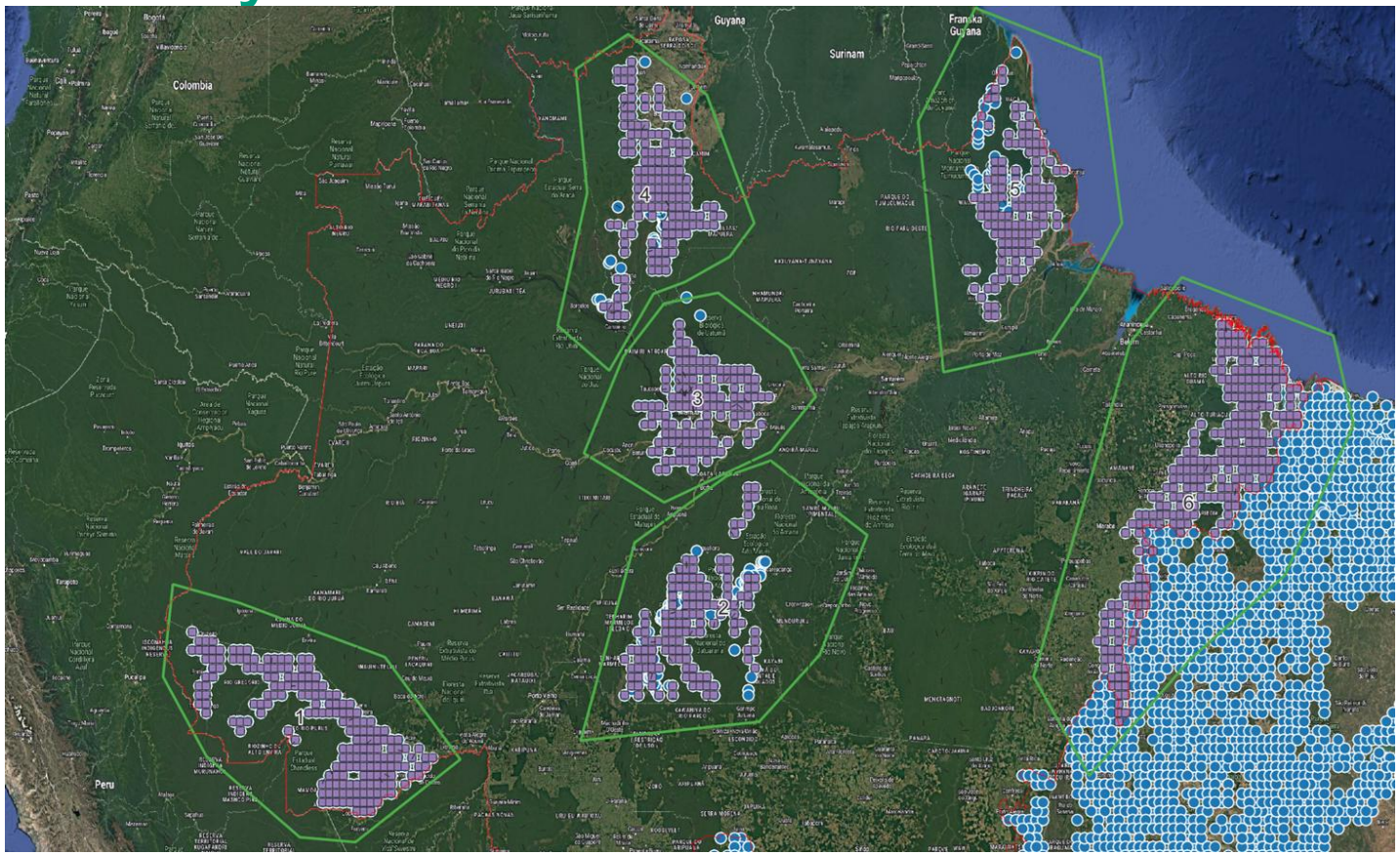
- Data for more than 10.000 plots with four subplots each of the Brazilian NFI



- Quality control
  - Comparing with data used in the previous ESA project S14Amazonas



Quality control

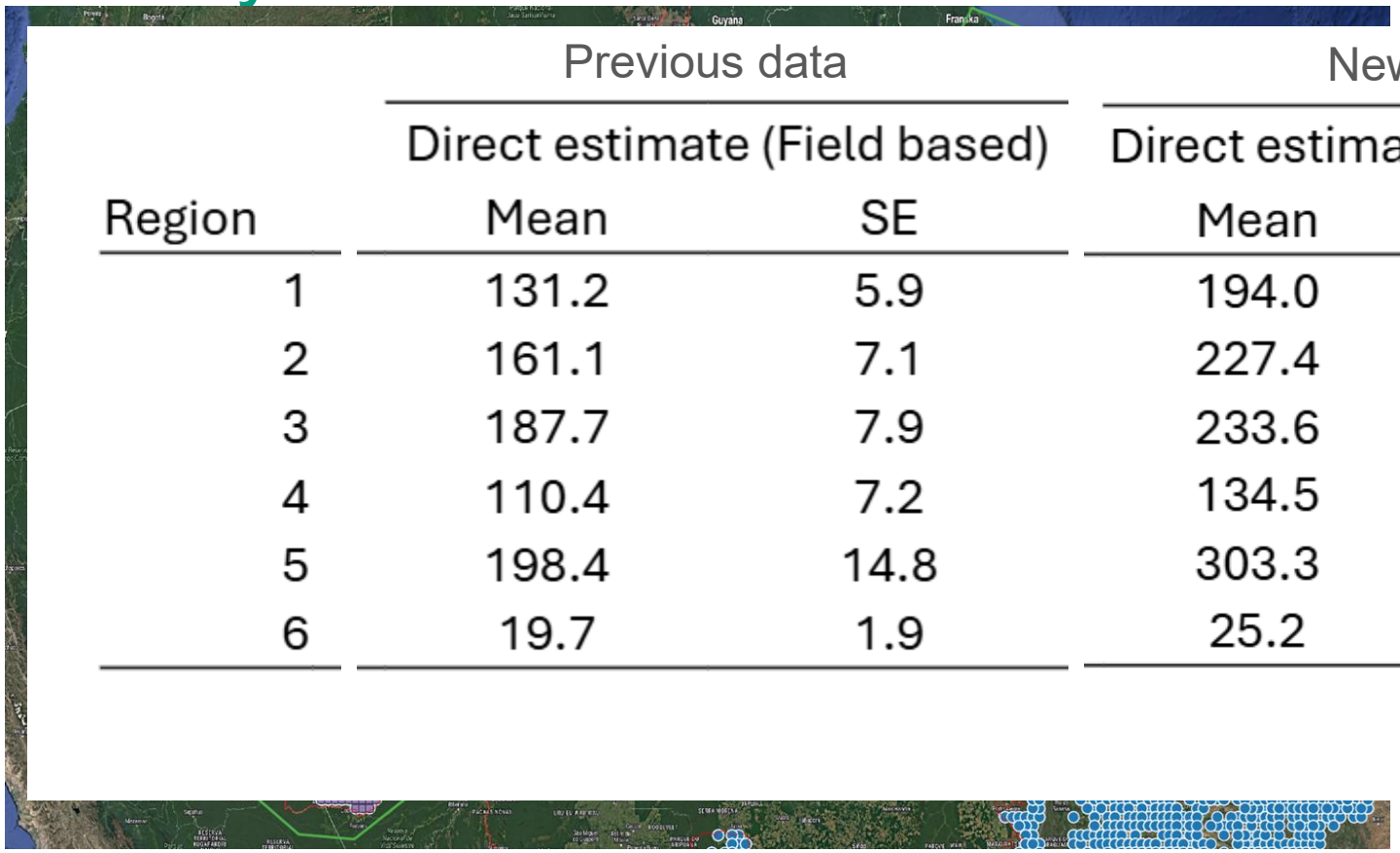


Region	Area (km^2)	n
1	76 000	190
2	54 000	135
3	52 800	132
4	58 800	147
5	50 000	125
6	104 400	261
	396 000	990



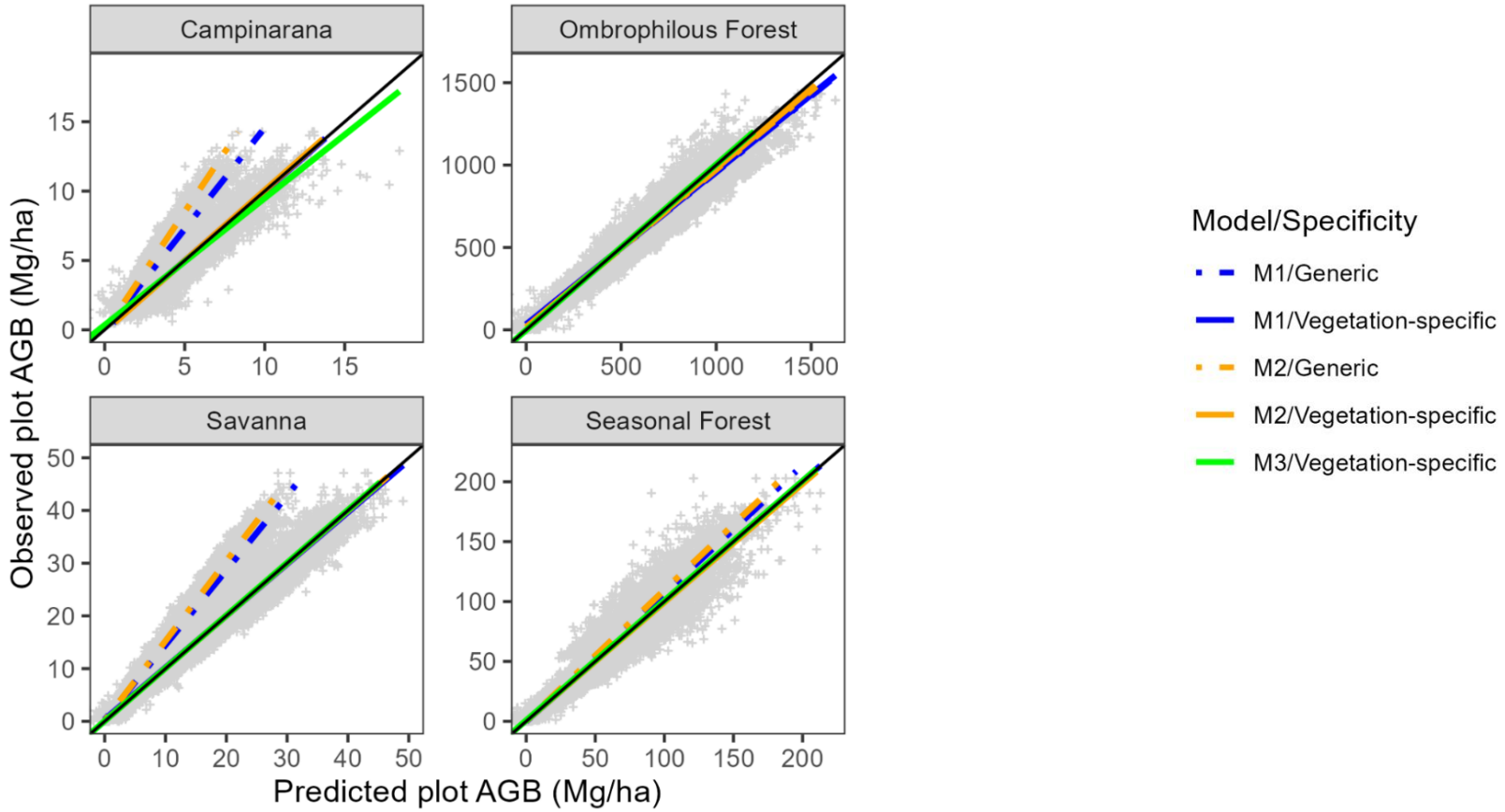
357 592 km^2

Quality control



Region	Previous data: Direct estimate (Field based) Mean	Previous data: Direct estimate (Field based) SE	New data: Direct estimate (Field based) Mean	New data: Direct estimate (Field based) SE	Relative difference (%)
1	131.2	5.9	194.0	8.7	+47.9
2	161.1	7.1	227.4	9.9	+41.2
3	187.7	7.9	233.6	9.7	+24.5
4	110.4	7.2	134.5	9.0	+21.8
5	198.4	14.8	303.3	20.7	+52.9
6	19.7	1.9	25.2	2.5	+27.9

Generic vs vegetation-specific biomass models



Preliminary results

					Model-assisted						
					BIOMASS						
			Direct estimate (Field based)		Original form					Calibrated	
Region	Area (km2)	n	Mean	SE	Mean	SE	D	Sign	D(%)	Mean	SE
AmazÃ´nia	895 200	2238	118.6	3.1							
Caatinga	634 400	1586	8.7	0.3							
Cerrado	1 468 800	3672	17.5	0.4							
Mata AtlÃ¢ntica	597 600	1494	47.2	1.8							
Pampa	185 600	464	18.3	1.9							
Pantanal	4 800	12	21.0	10.1							
	3 786 400	9466									

Relative efficiency:

$$\mathrm{RE} = \frac{\hat{V}_{\mathrm{Field}}}{\hat{V}_{\mathrm{Field+EO-based\ maps}}}$$

Relative efficiency for mean estimates of AGB

	BIOMASS	
Region	Original form	Calibrated
AmazÃ´nia		
Caatinga		
Cerrado		
Mata AtlÃ¢ntica		
Pampa		
Pantanal		